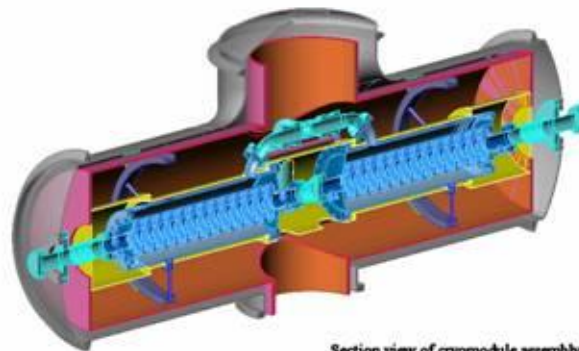


Section Views



Section view of cryomodule assembly.



TOPICS

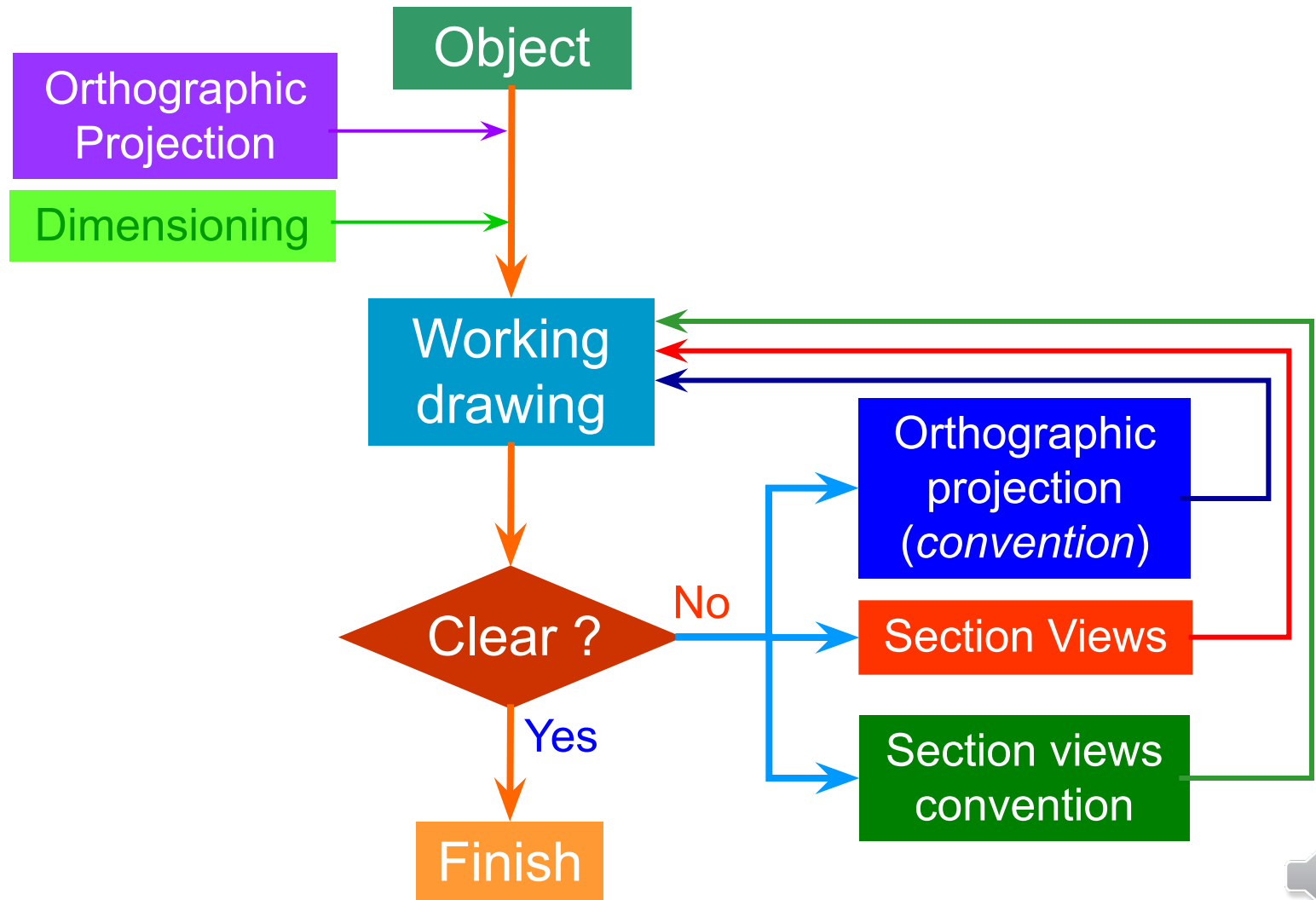
- Introduction
- Terminology & common practices
- Kind of sections
- Dimensioning



Introduction



GRAPHICS COMMUNICATION WITH ENGINEERING DRAWING



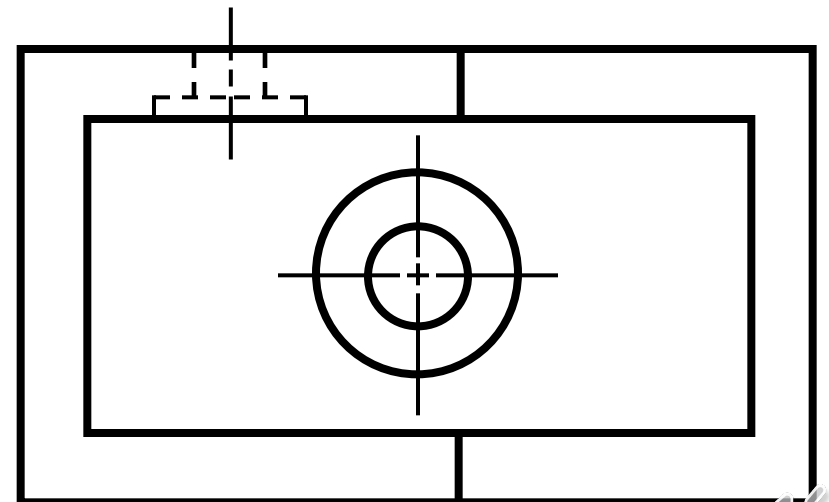
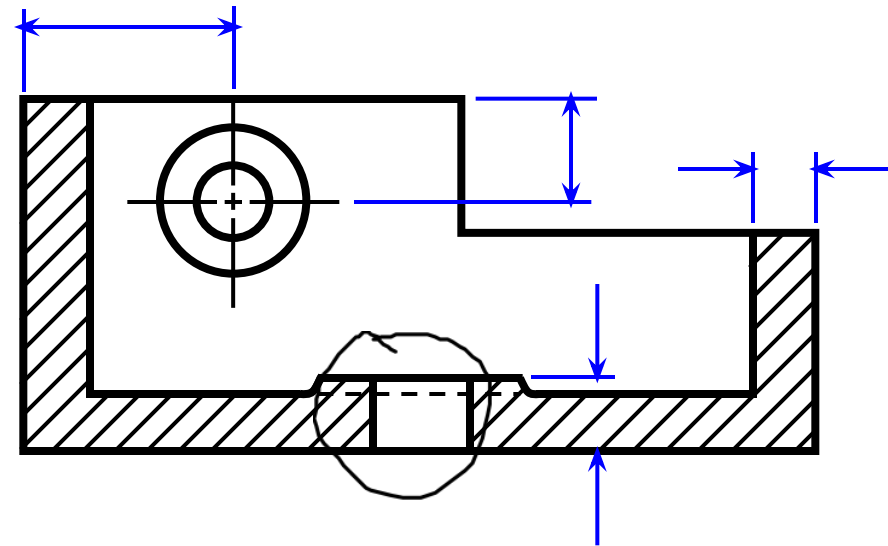
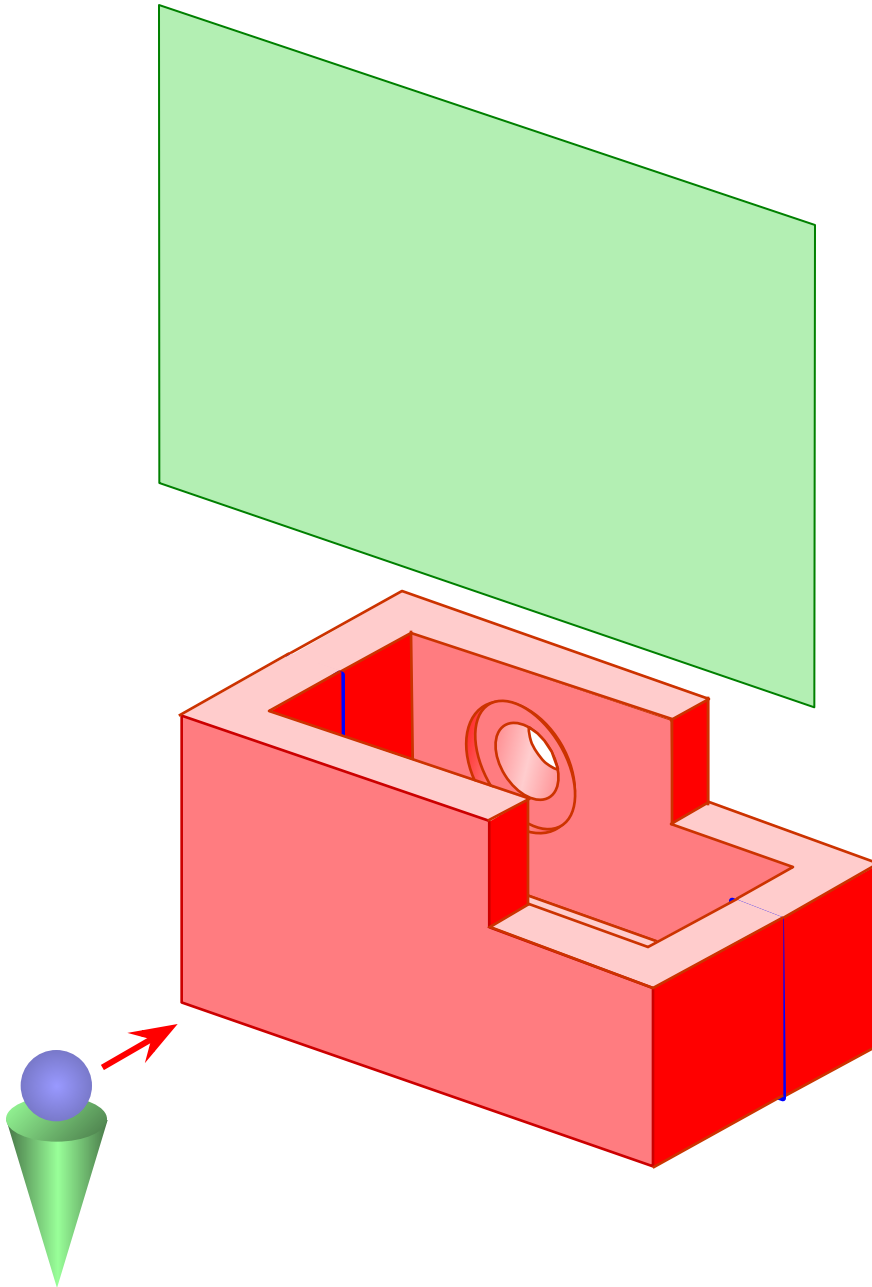
PURPOSES OF SECTION VIEWS

- Clarify the views by
 - ❖ reducing or eliminating the hidden lines.
 - ❖ revealing the cross sectional's shape.
- Facilitate the dimensioning.

Let See the



EXAMPLE : Advantage of using a section view.

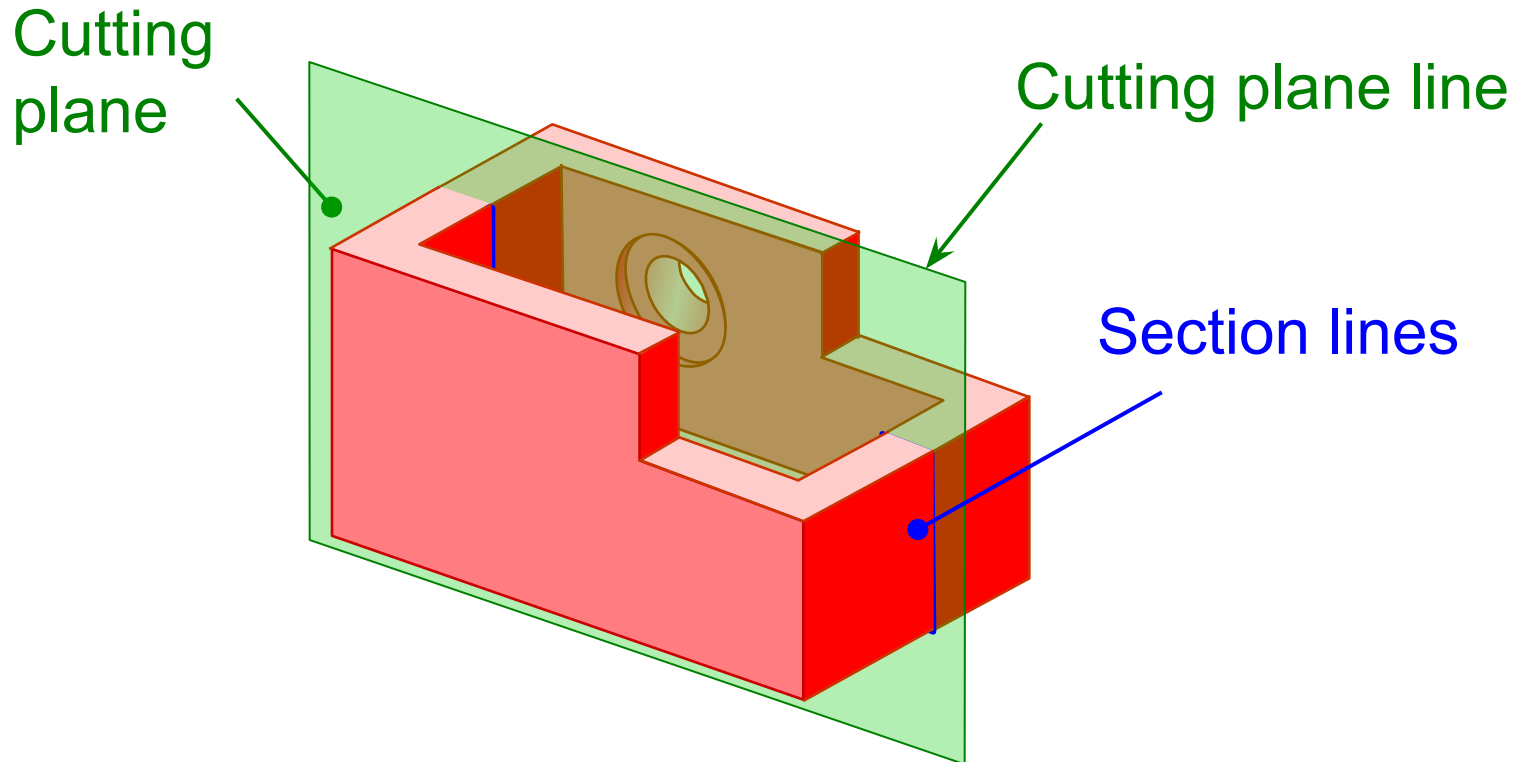


Terminology and common practices



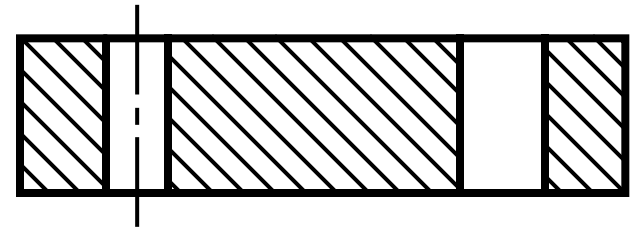
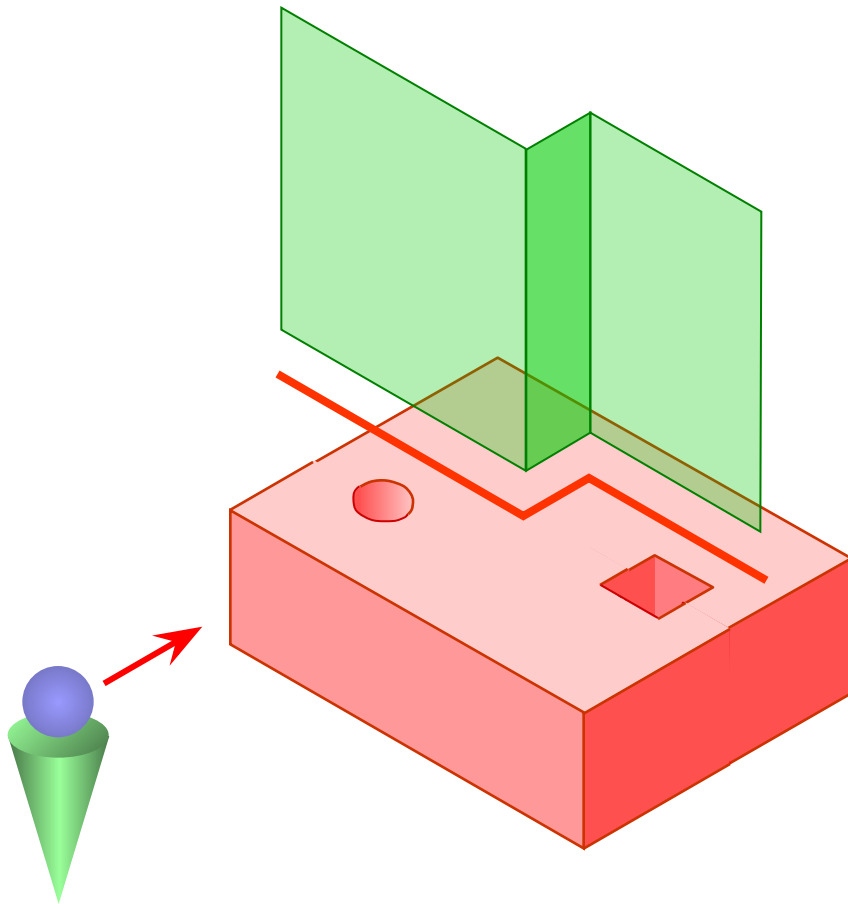
CUTTING PLANE

Cutting plane is a plane that ***imaginarily cuts*** the object to reveal the internal features.

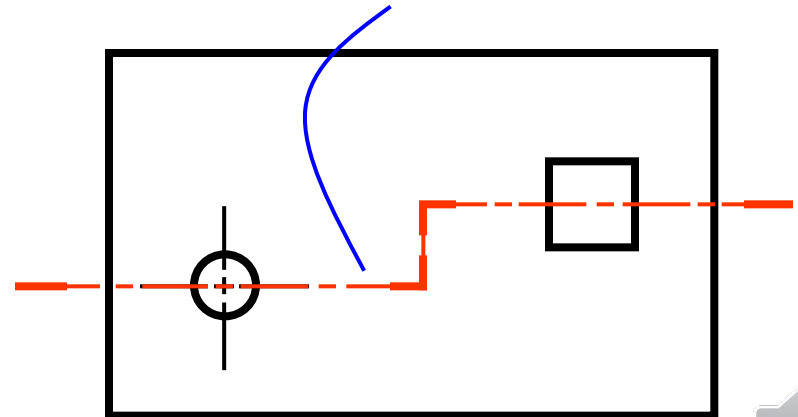


CUTTING PLANE LINE

Cutting plane line is an ***edge view*** of the cutting plane.

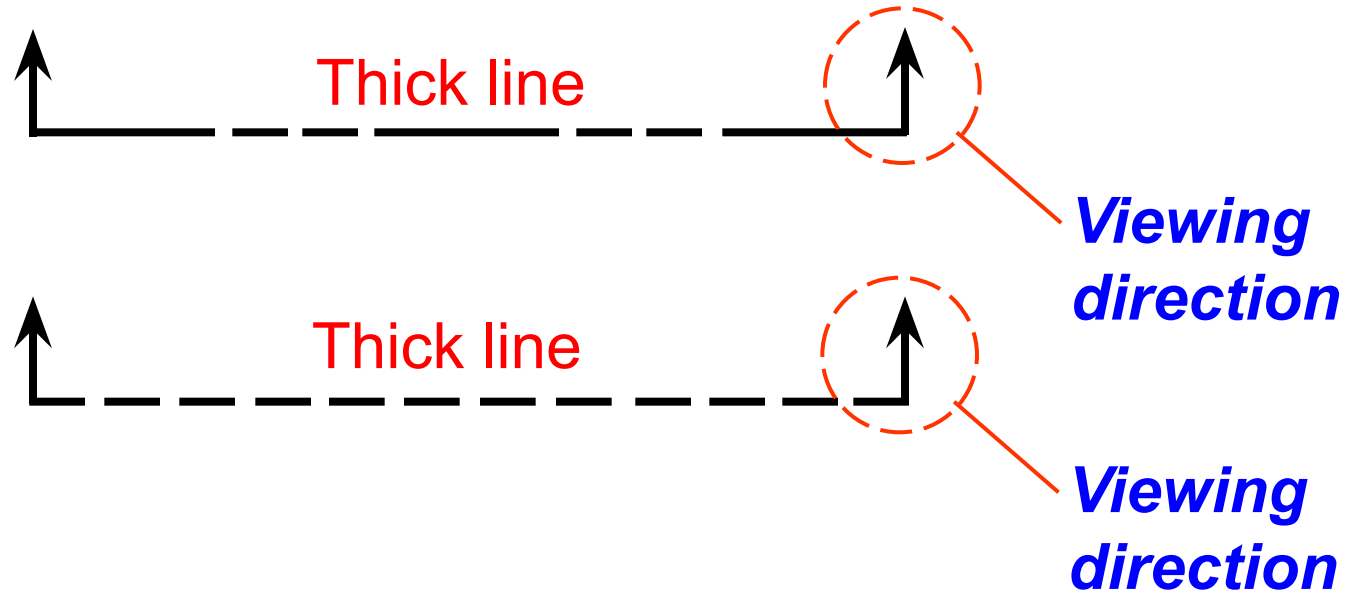


Indicate the ***path*** of cutting plane.

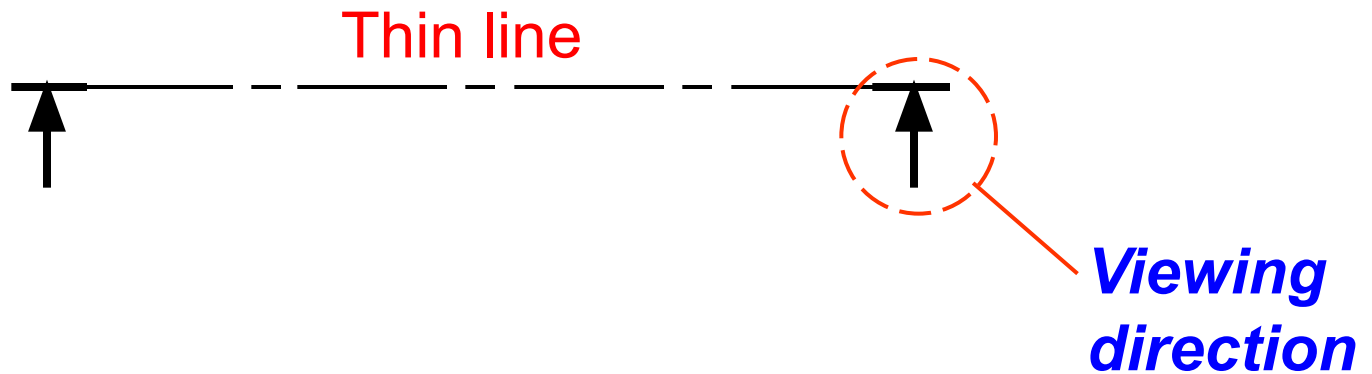


CUTTING PLANE LINESTYLES

ANSI
standard

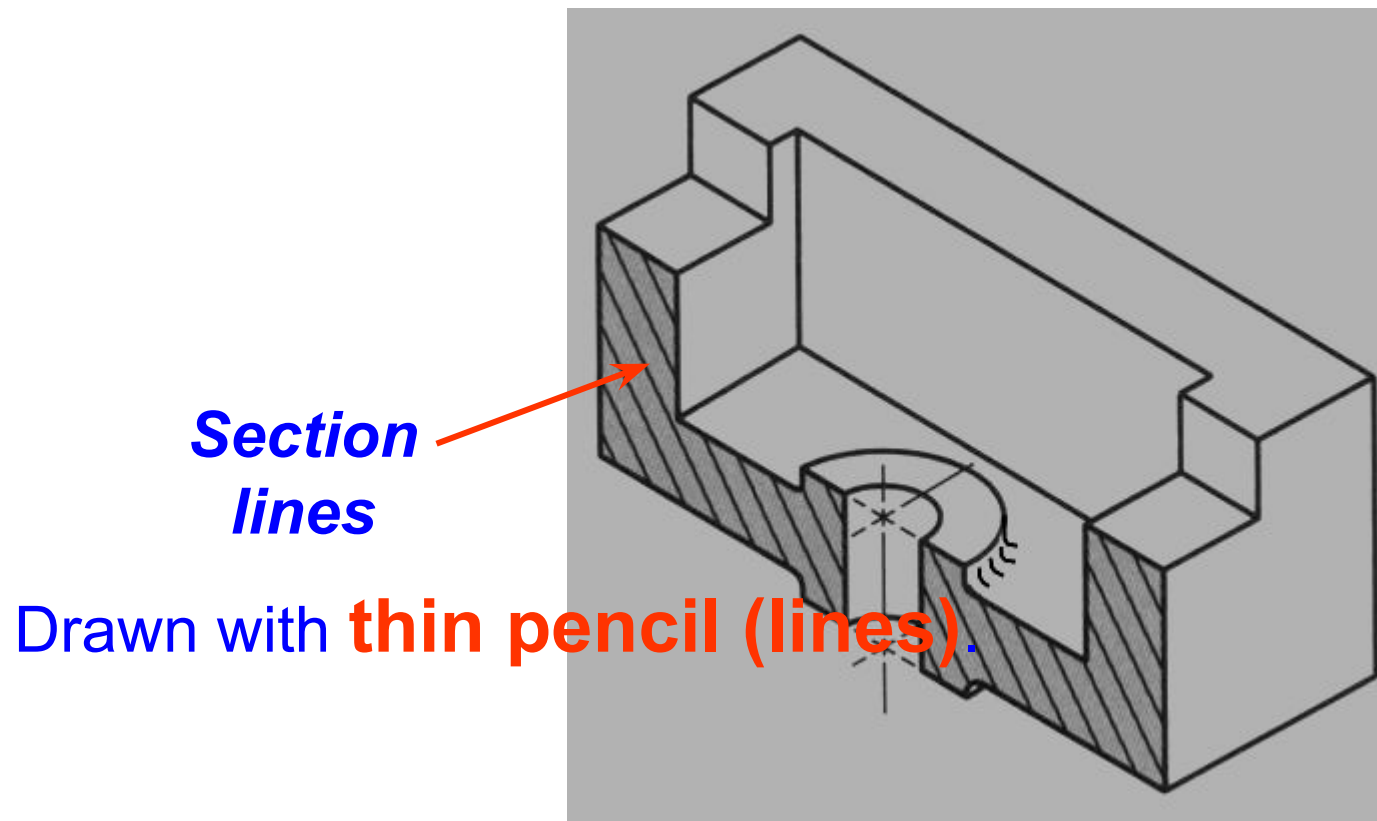


TS & ISO
standard



SECTION LINING

Section lines or **cross-hatch lines** are used to *indicate the surfaces that are cut by the cutting plane.*



SECTION LINES SYMBOLS

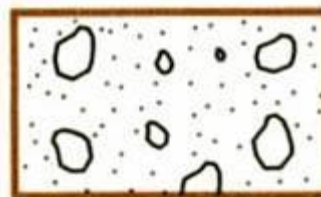
- The section lines are different for each of material's type.
- For practical purpose, the cast iron symbol is used most often for any materials.



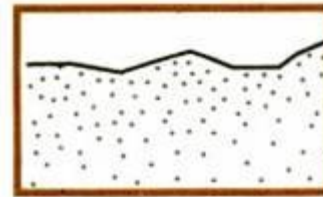
Cast iron,
Malleable iron



Steel



Concrete



Sand

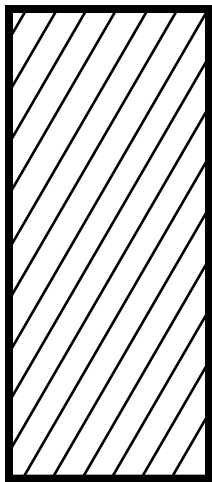


Wood

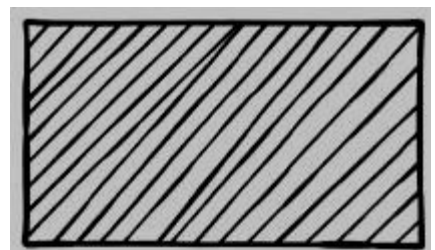
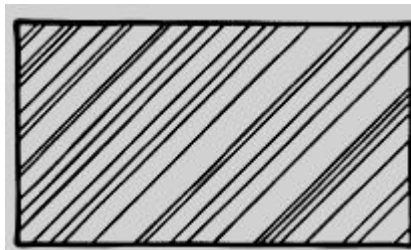
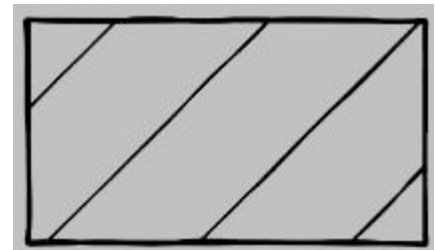
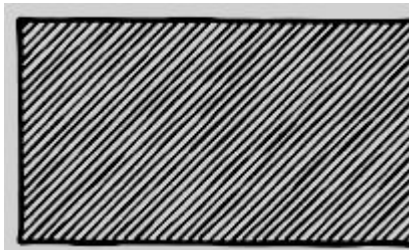


SECTION LINING PRACTICE

- The spaces between lines may vary from 1.5 mm for small sections to 3 mm for large sections.



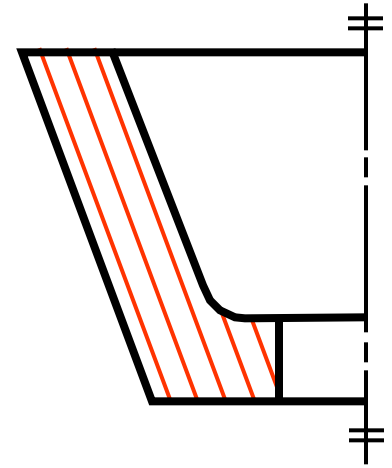
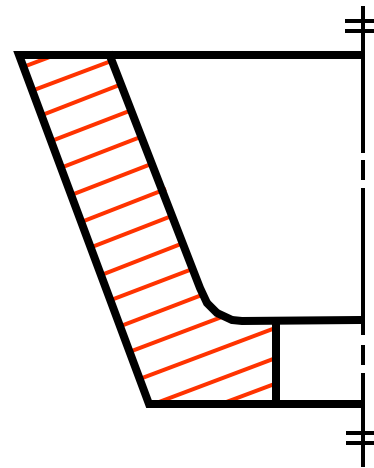
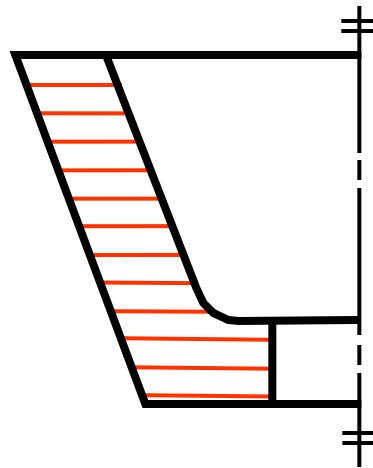
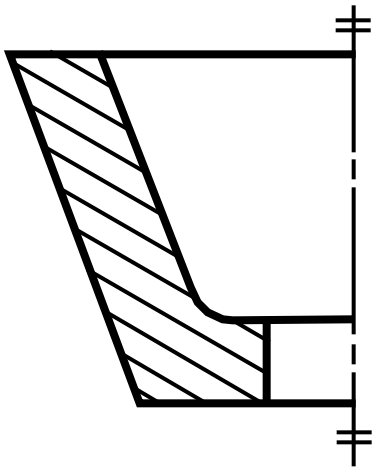
COMMON MISTAKE



SECTION LINING PRACTICE

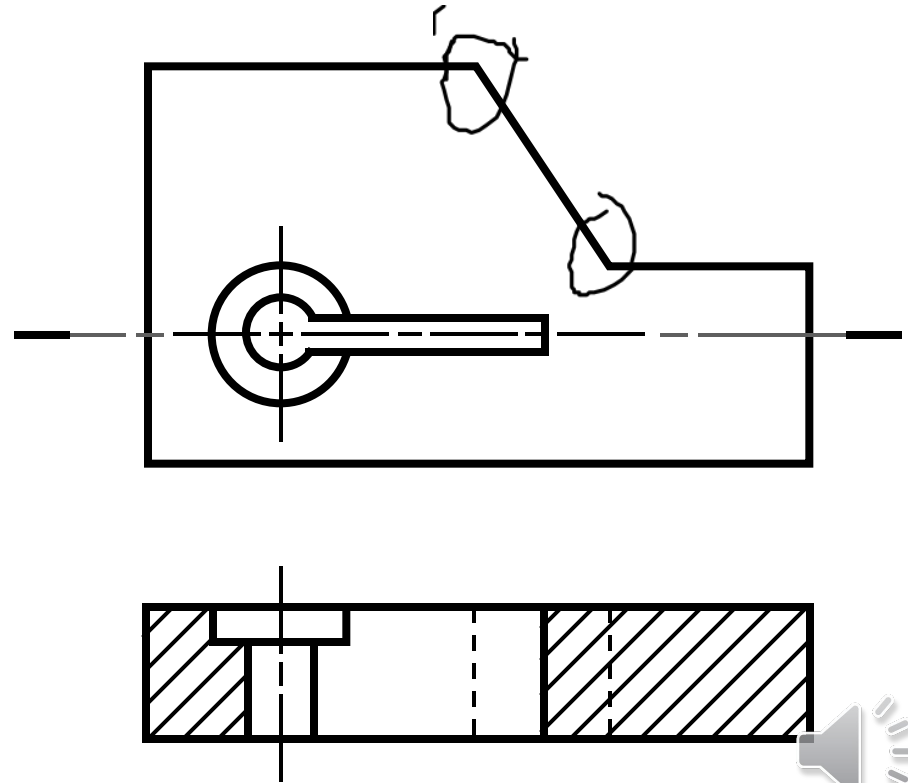
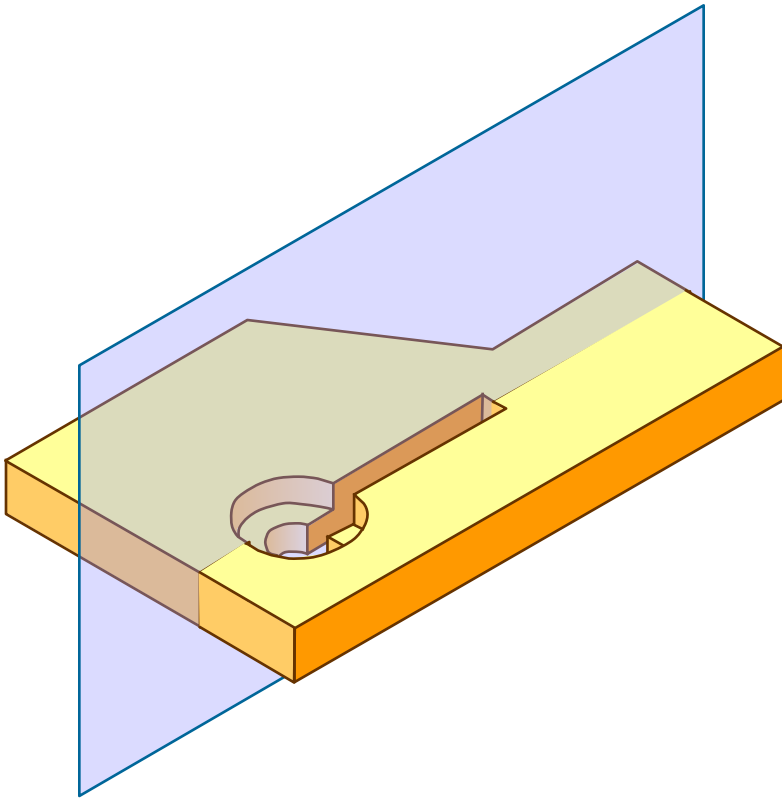
- It ***should not*** be drawn *parallel* or *perpendicular* to contour of the view.

COMMON MISTAKE



TREATMENT OF HIDDEN LINES

- Hidden lines are *normally omitted* from section views.



Kinds of Sections



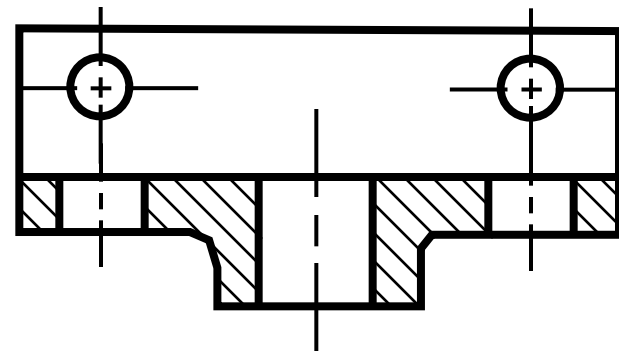
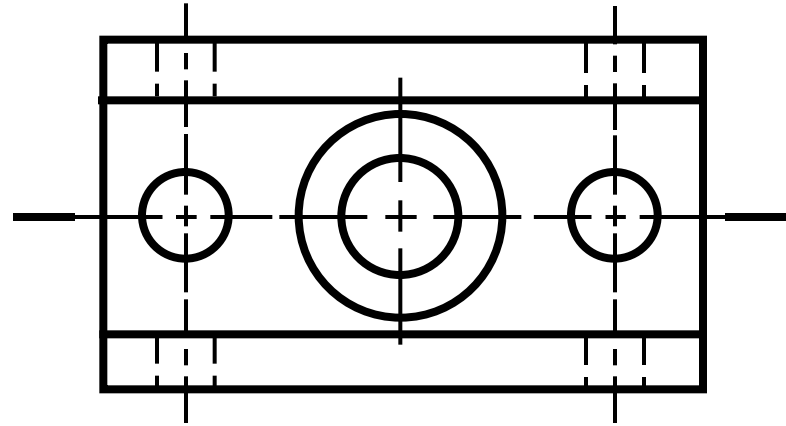
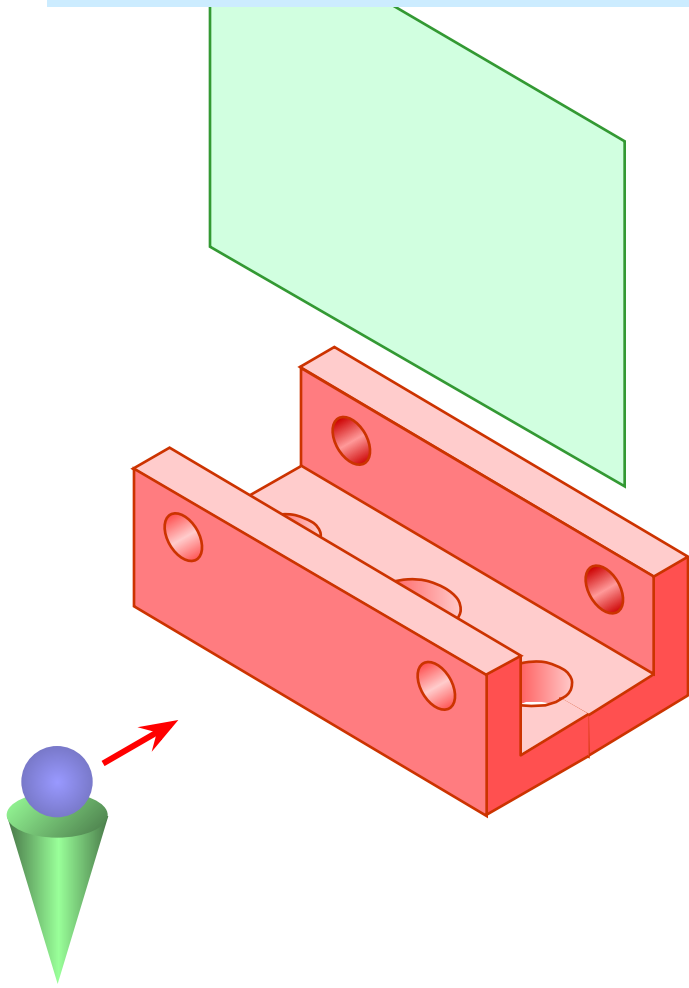
KIND OF SECTIONS

1. Full section
2. Offset section
3. Half section
4. Broken-out section
5. Revolved section (aligned section)
6. Removed section (detailed section)



FULL SECTION VIEW

The view is made by passing the *straight* cutting plane *completely through* the part.



OFFSET SECTION VIEW

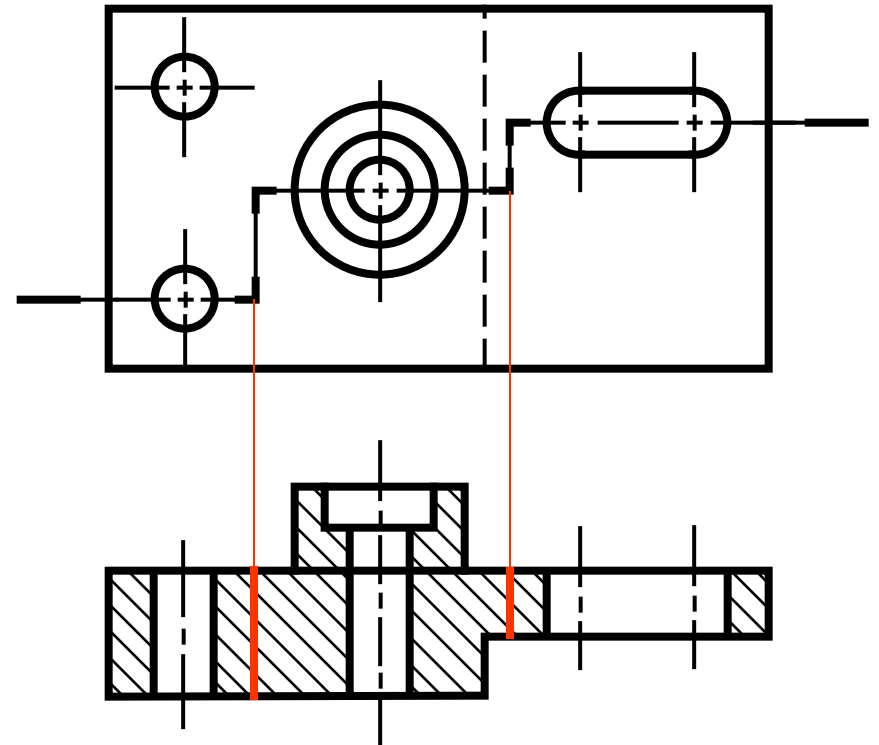
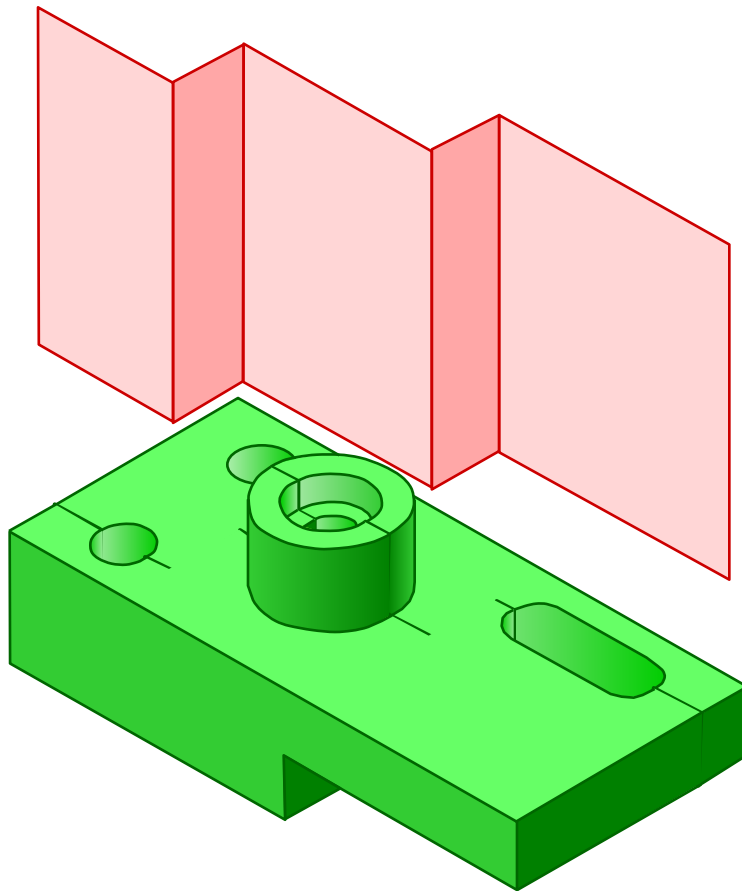
The cutting plane is *off-set* to include features that are not in a straight line.

It is possible for the cutting plane to *change directions*, to *minimise on the number* of sectional views required to capture the necessary details.

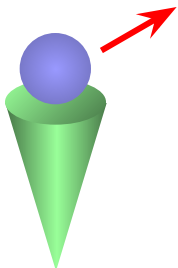


OFFSET SECTION VIEW

The view is made by passing the *bended* cutting plane *completely through* the part.



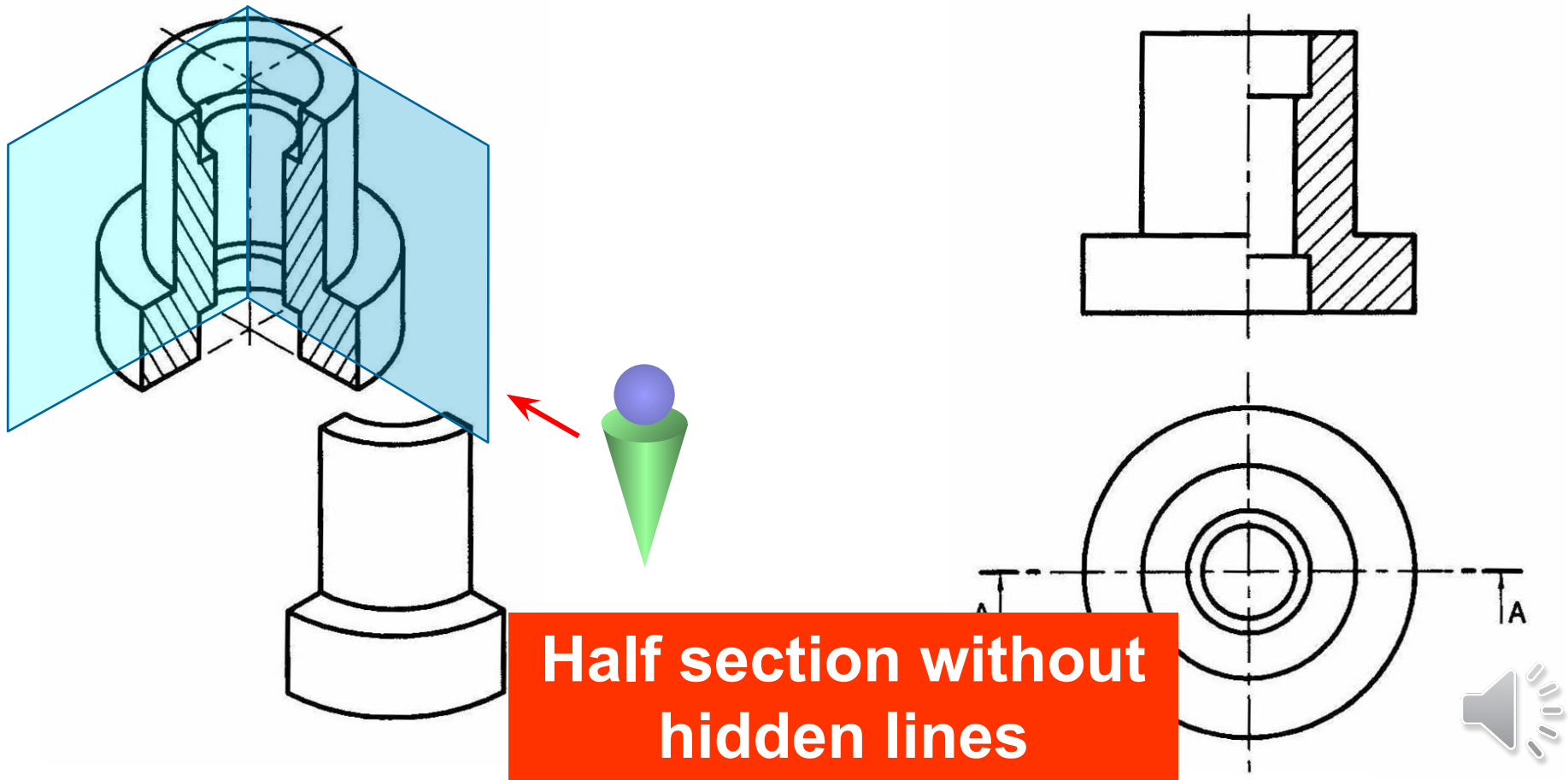
Do not show the edge views of the cutting plane.



HALF SECTION VIEW

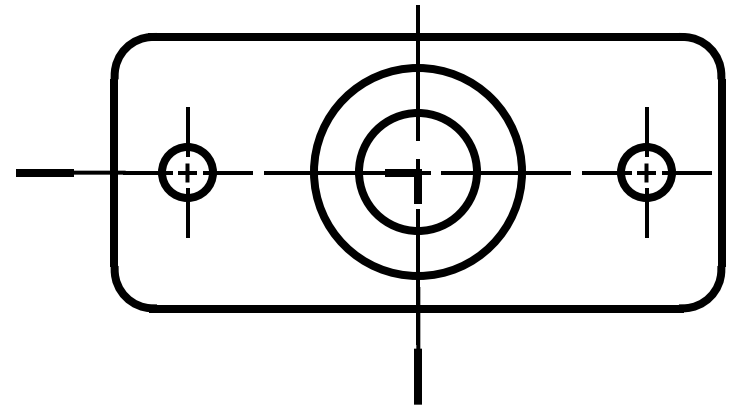
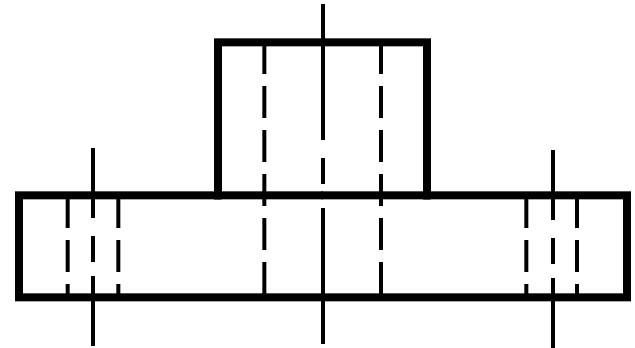
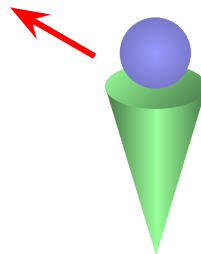
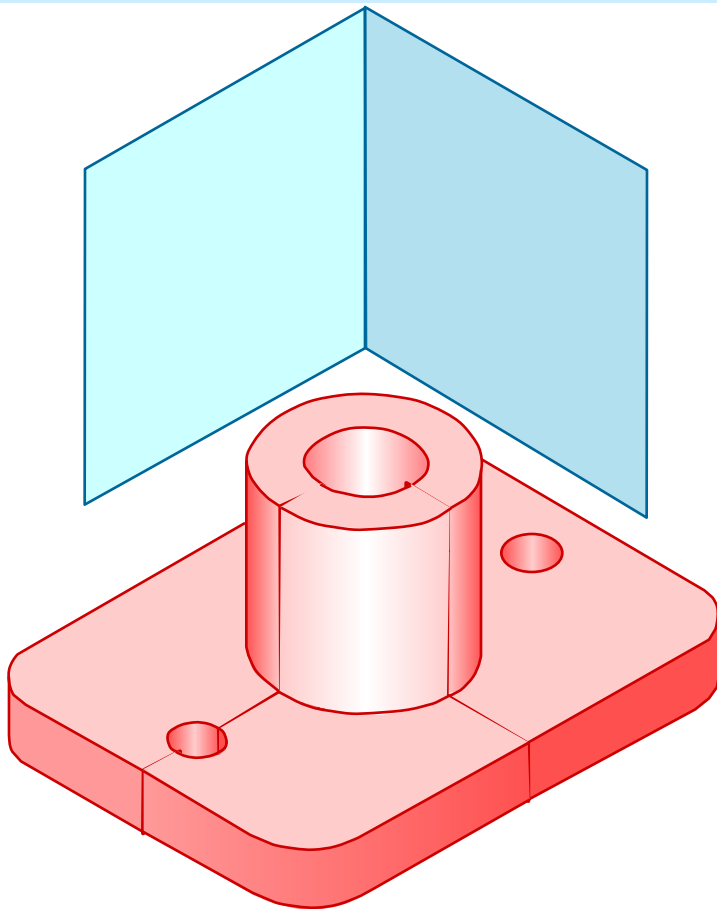
A half-section is a view of an object showing *one-half* of the view in section.

Symmetrical parts can be shown in half sections.



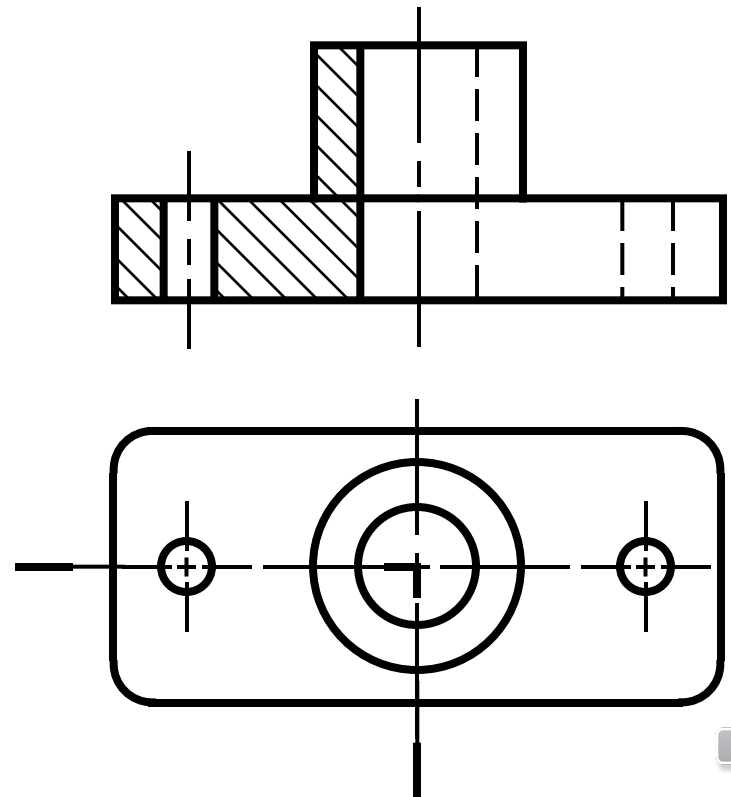
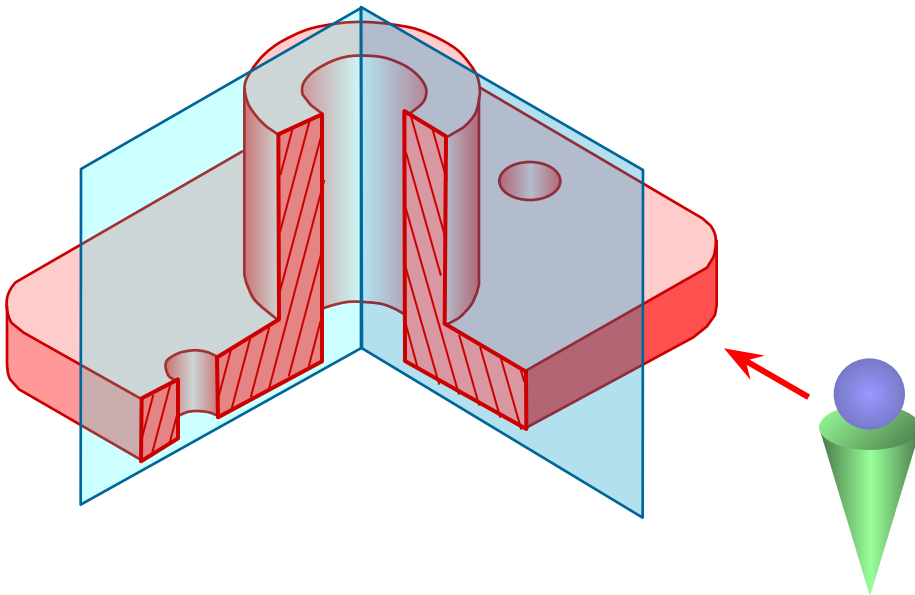
HALF SECTION VIEW

The view is made by passing the cutting plane *halfway* through an object and remove a *quarter* of it.



HALF SECTION VIEW

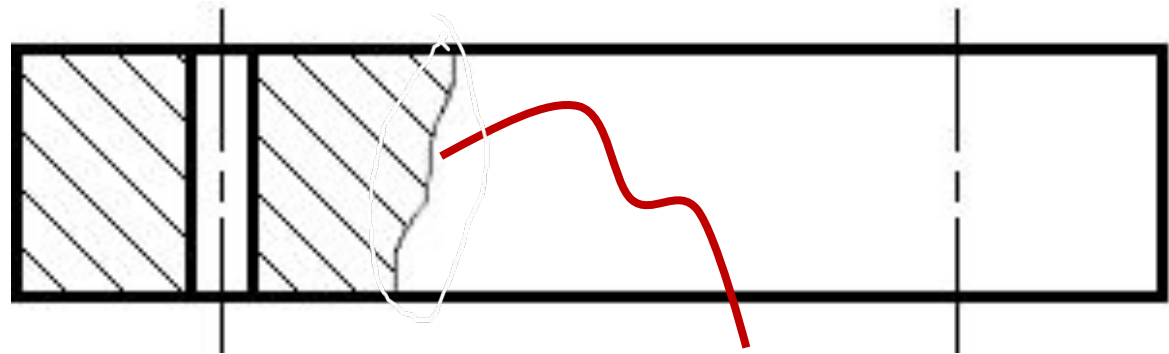
- A **center line** is used to separate the sectioned half from the unsectioned half of the view.
- **Hidden line** is omitted in unsection half of the view.



BROKEN-OUT SECTION VIEW

It is common practice to section a part of an object when only small areas need to be sectioned to indicate the important details

The example shows a **broken-out** sectional view to indicate a through-hole in a plate.

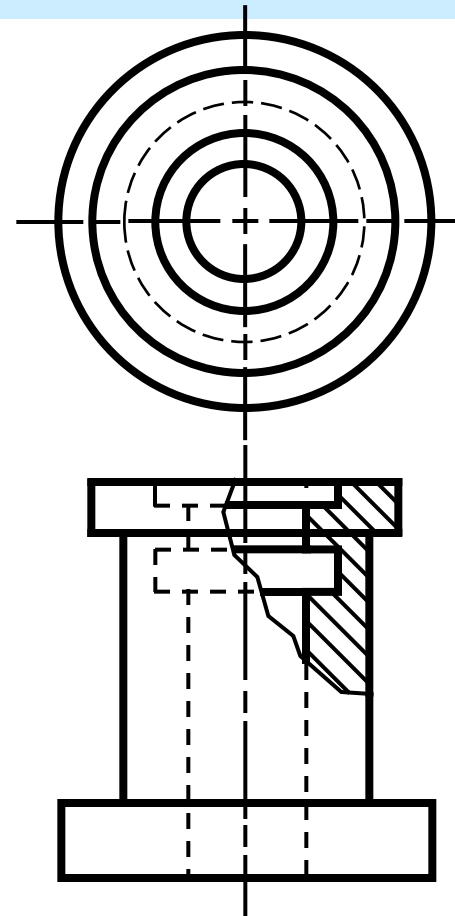
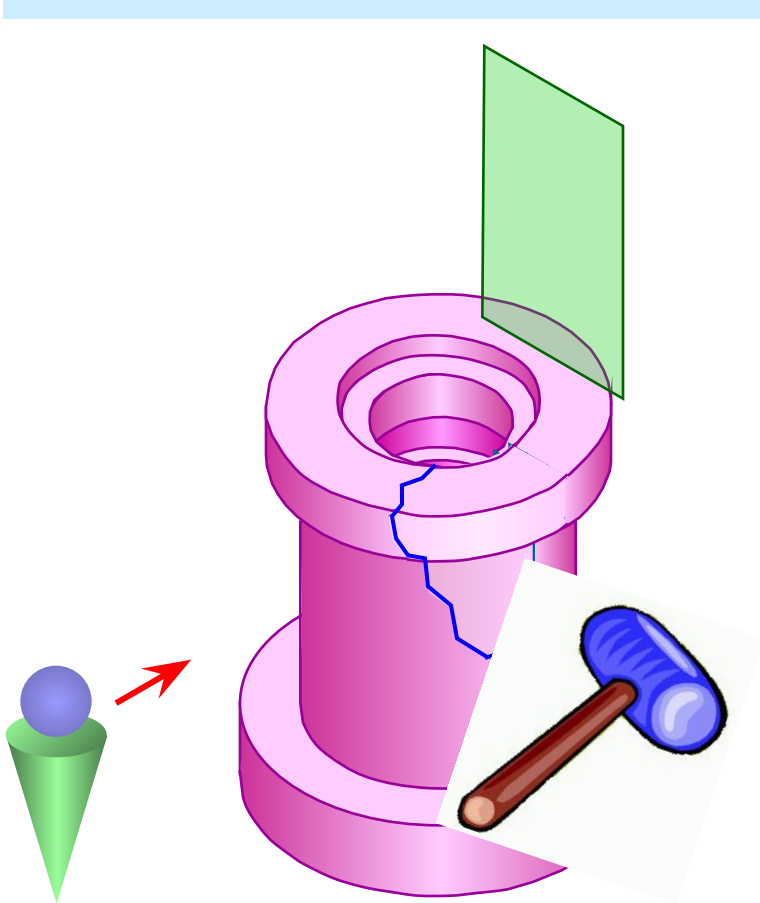


The lines indicating the end of the section is a thin continuous line



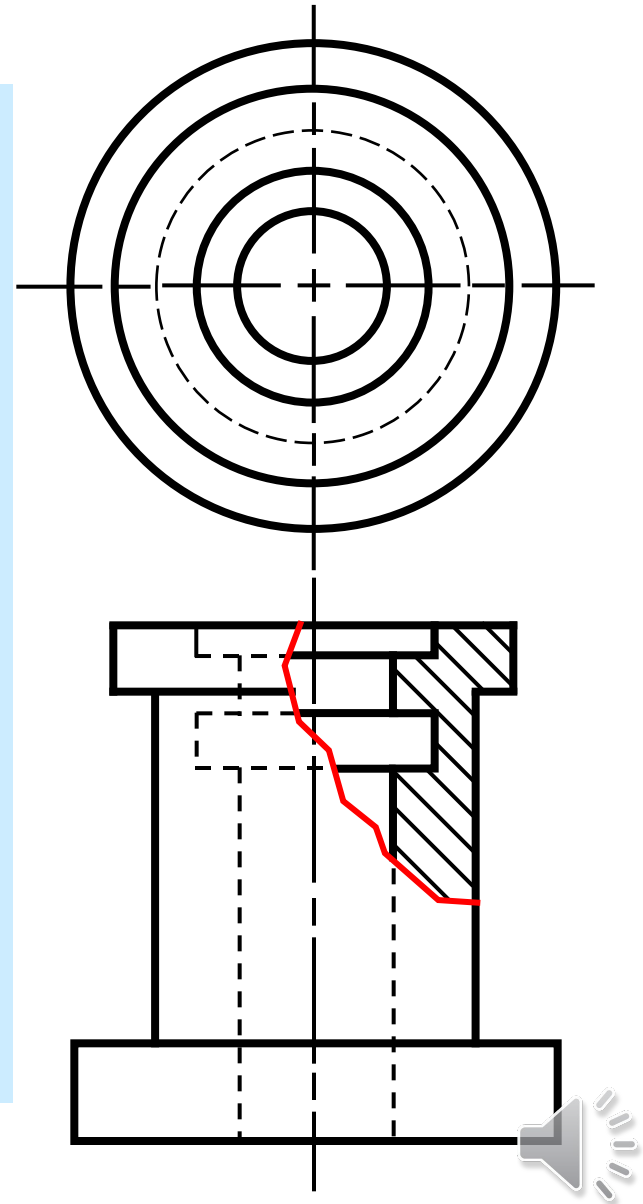
BROKEN-OUT SECTION VIEW

The view is made by passing the cutting plane normal to the viewing direction and removing the portion of an object in front of it.

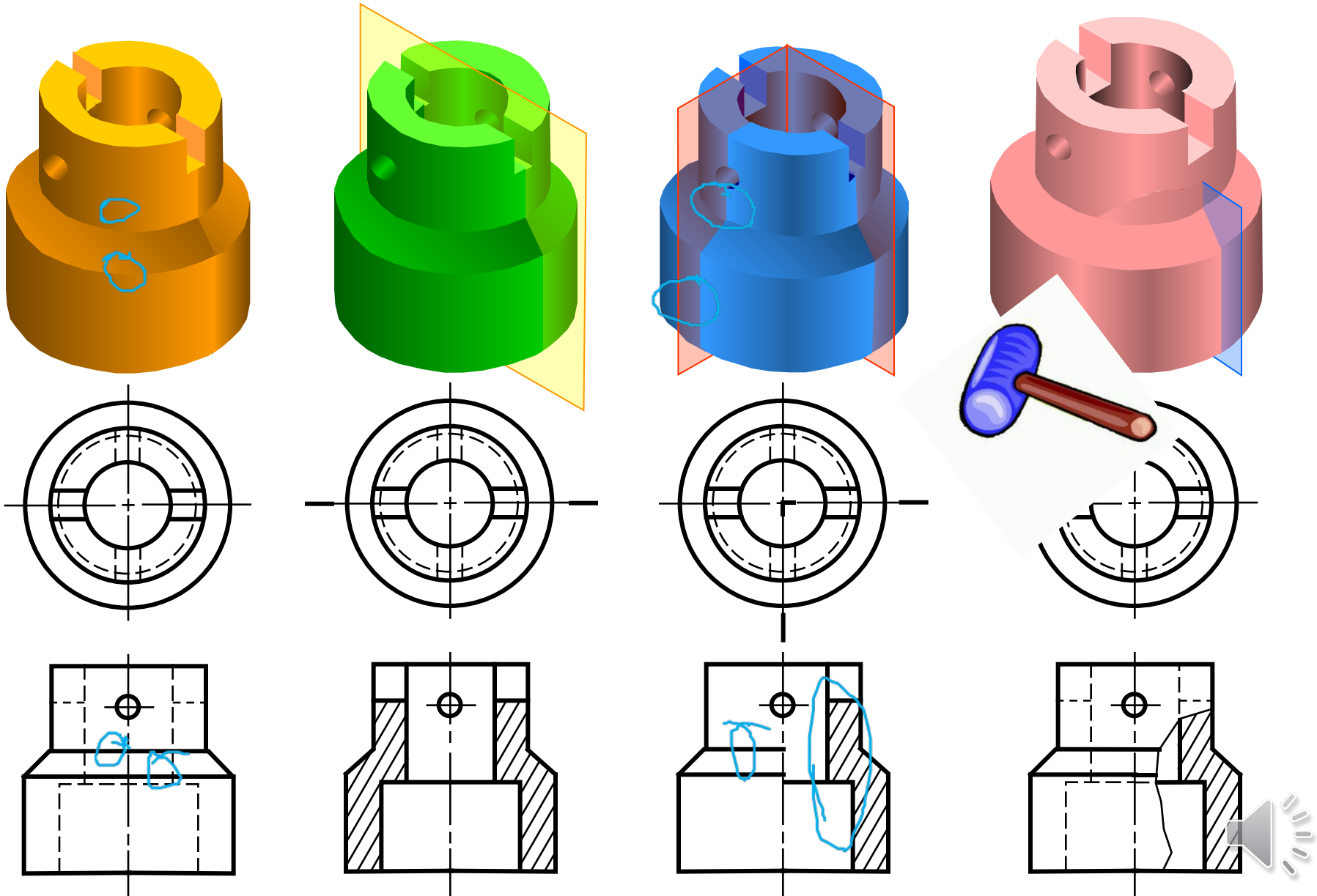


BROKEN-OUT SECTION VIEW

- A **break line** is used to separate the sectioned portion from the unsectioned portion of the view.
- Break line is a thin continuous line (**0.25**) and is drawn freehand.
- There is **no** cutting plane line.



EXAMPLE : Comparison among several section techniques

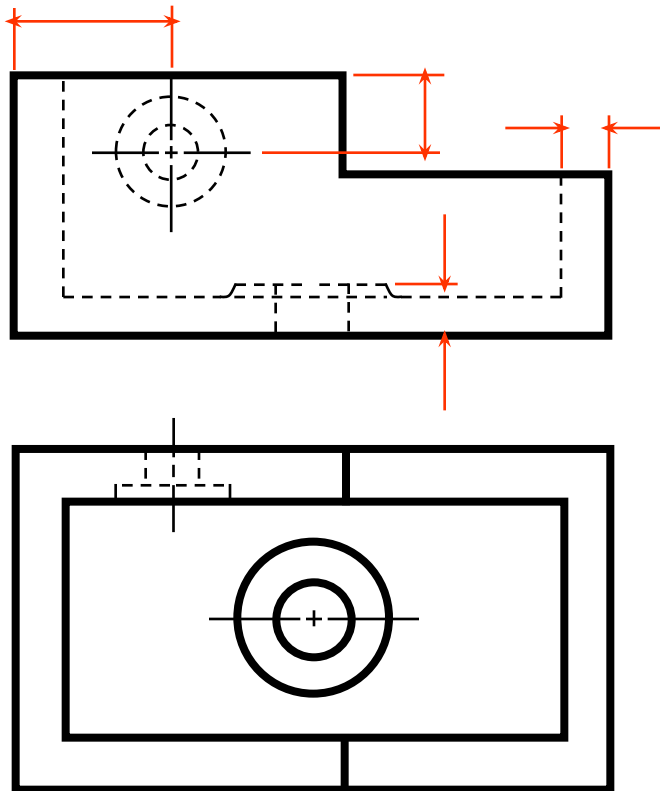


Dimensioning in Section View

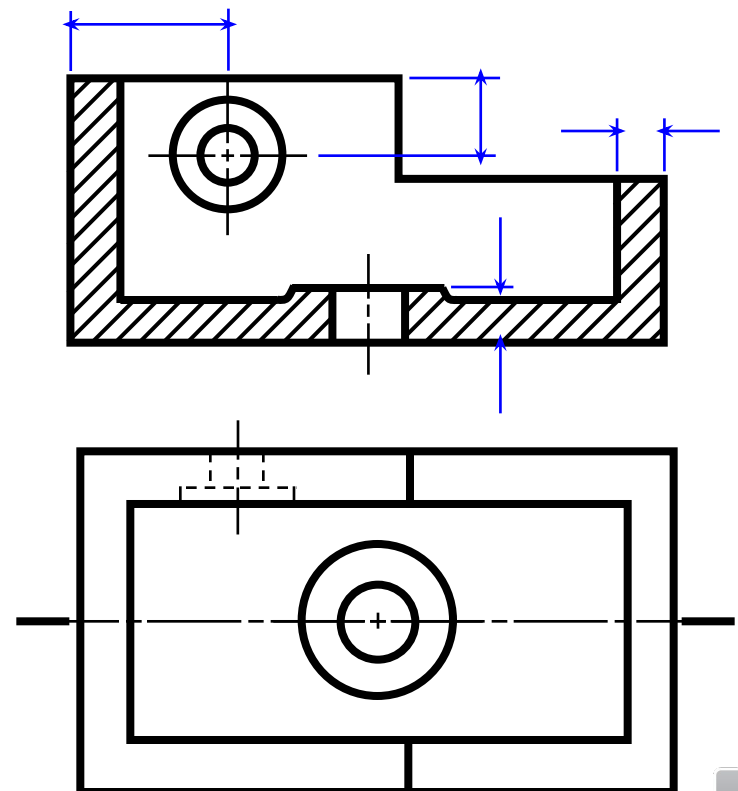


- In most cases, dimensioning of the section views follows the typical rules of dimensioning.

POOR

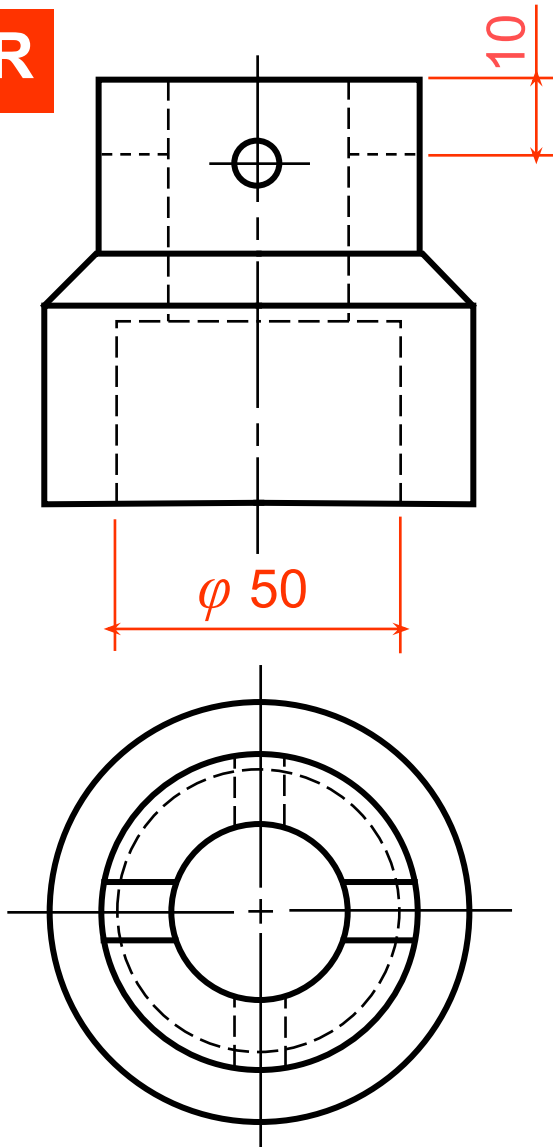


GOOD



DIMENSIONING

POOR



GOOD

